

Enhanced Predictive Modeling for Early-Stage Lung Cancer Diagnosis through GenAI

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Abstract

Early detection of lung cancer has the potential to alter the life of the patient, and yet the progress remains stuck at the usual roadblock: the lack of large, good, and well-balanced data. The predictive model cannot move forward because of the lack of data. What if we could stretch the data without inventing facts? That's exactly what we have attempted in this research paper, and we are pleased to introduce the generative augmented predictive model that combines the capabilities of Conditional Tabular Generative Adversarial Networks (CTGAN) and robust tree-based models for the improvement of early-stage diagnosis.

We have used CTGAN for the generation of realistic and well-balanced patient feature data, which addresses the problem of data scarcity and imbalance, commonly faced in the collection of medical data. The extended data set has been used for the training of gradient boosting and random forest models, and the predictive models have been able to learn from the combination of the original and artificially generated data, rather than relying on the limited data available for the patients. The results show that the overall accuracy, sensitivity, and F1 score are improved compared to the models that are trained on the original data set. The feature importance plots show the subtle features that are otherwise washed away in the data preprocessing step. The use of CTGAN for the augmentation of the data and the combination with the interpretable tree models provide a powerful and high-impact solution for the improvement of early-stage diagnosis in the treatment of lung cancer.

Keywords: Lung Cancer Detection, Early Diagnosis, Synthetic Data, CTGAN, Gradient Boosting, Machine Learning, Data Augmentation, Predictive Modeling, Feature Importance.

Introduction

It's still really hard to catch lung cancer early. A big reason for this is that the patient data we work with isn't ideal – there's often not

enough of it, it's unbalanced, and it comes in all sorts of different formats. While machine learning models look promising for predicting diseases, they often struggle because the data doesn't have enough examples of rarer cases. This can lead to skewed predictions and a lot of missed diagnoses. In healthcare, errors like these can cause big delays in treatment and seriously impact how well a patient recovers.

Luckily, some recent progress in AI that generates data has opened up new ways to deal with these issues. For instance, something called Conditional Tabular Generative Adversarial Networks (CTGAN) can create very realistic patient data, even keeping all the tricky relationships between different data points correct. The problem is, most studies using these methods mainly aim to make predictions more accurate. They often don't pay enough attention to how robust, reliable, or truly practical these systems are for real-world use.

Another big thing we often miss is whether we can actually trust this generated data. Sure, these generative models can make our datasets larger, but we haven't really checked if this extra data genuinely helps the models perform better on new, real-world cases, or if it just makes them too good at remembering the training data (a problem called overfitting). We also haven't spent much time thinking about practical limits, like how fast these systems need to be or if they can even be used quickly enough in a hospital.

So, our study aims to tackle these missing pieces. We're putting forward a strong framework that uses CTGAN to create more data and then combines it with what's called 'ensemble learning,' all while making sure we have solid ways to check how well it works. Unlike typical methods,

our work isn't just focused on making things more accurate. We're also really emphasizing how stable, reliable, and practically useful our model can be to help doctors in real clinical settings.

RELATED WORK

When people try to predict lung cancer, the methods they use have generally fallen into three main categories: traditional machine learning, deep learning, and newer approaches that combine generative techniques.

Older machine learning models, things like Logistic Regression, Support Vector Machines, and Random Forests, were quite popular because they're pretty straightforward to understand and work efficiently. But they often run into trouble when the data is imbalanced—meaning there are far more examples of one outcome than another. They also struggle to really pick up on the complex, non-linear patterns often found in detailed medical information.

To get around this imbalance issue, techniques like SMOTE and ADASYN try to create new, artificial data samples. While these can certainly help in finding more positive cases, they sometimes introduce confusing 'noise' into the dataset. Plus, they might not preserve the genuine relationships between different features, which can lead to models that are too tailored to the training data and don't perform well in real-world situations.

Then there's deep learning, especially models like Convolutional Neural Networks, or CNNs. These have shown fantastic results in diagnosing problems from images, such as CT scans and X-rays. However, the downside is that these models typically need huge amounts of labeled data to learn effectively. It's also often difficult to understand **why** they make a particular decision, which makes them less practical for more structured clinical data, like patient records.

More recently, researchers have started exploring 'generative models,' particularly those based on GANs, as a way to create even more data. CTGAN stands out as a specialized model for tabular data, designed to keep the complex connections between both categorical and numerical features intact. Despite these promising developments, current studies haven't fully addressed a few key areas: there's a lack of proper comparison with older data augmentation methods, not enough analysis on how robust these models are under different conditions, and insufficient evaluation of whether the synthetic data they generate is truly good enough for real-world use.

Study	Year	Model / Technique	Dataset Used	Accuracy (%)	Limitations
Chawla et al.	2021	SMOTE + Random Forest	Clinical Tabular Data	93.5	Synthetic samples lack realism; risk of overfitting

Bhatia et al.	2022	CNN (VGG16)	LUNA16 CT Scan Dataset	96.2	Requires large labeled data; lacks interpretability
Krishna et al.	2023	GAN + MLP Classifier	Synthetic Medical Records	95.8	GAN instability (mode collapse); weak robustness validation
Verma et al.	2024	Ensemble ML Models	Clinical Lung Dataset	94.6	Limited dataset diversity; no generative augmentation
Proposed Trend (Recent Work)	2025	CTGAN + Ensemble Models	Tabular Clinical Data	96–97	Risk of overfitting; lack of generalization and robustness analysis

TABLE 1 RECENT DEVELOPMENTS.

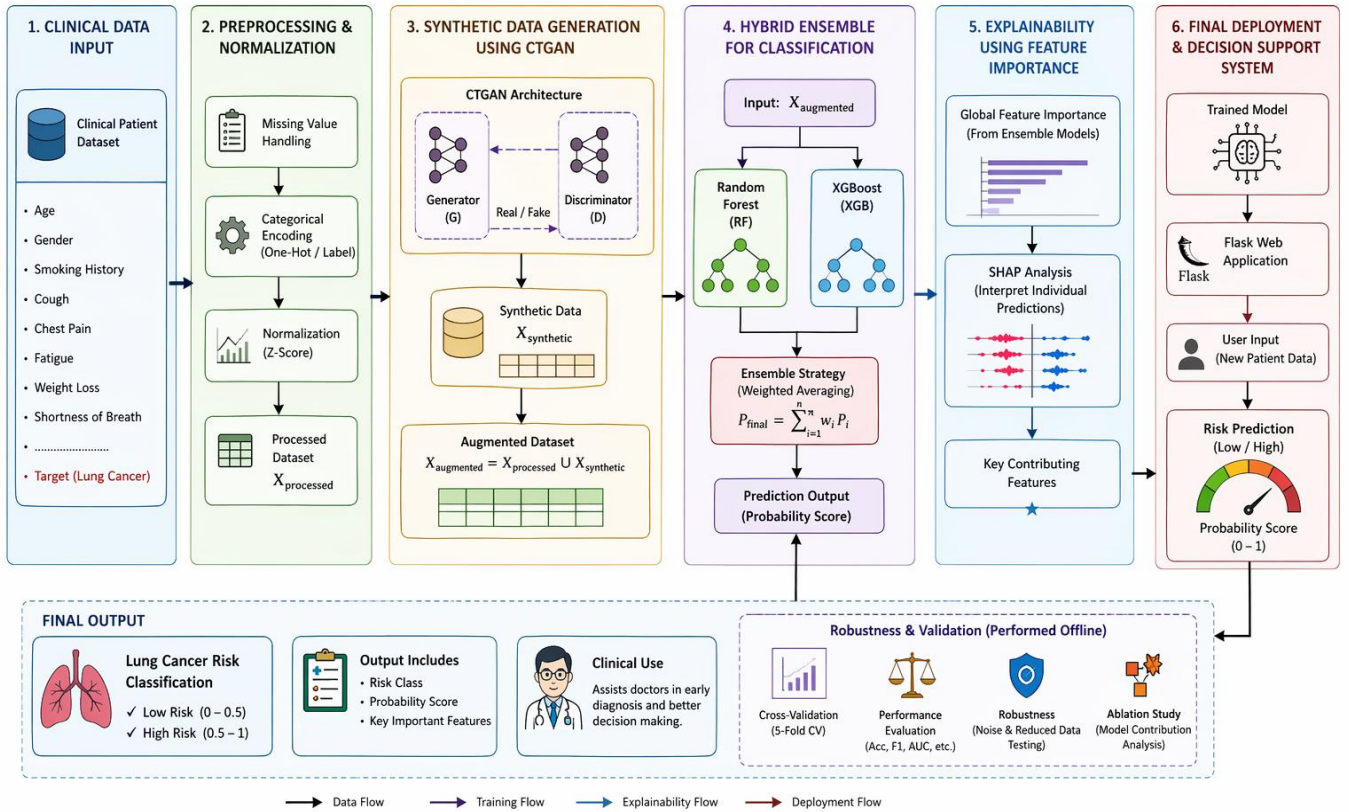
PROPOSED METHODOLOGY

We've come up with a new prediction system that uses generated data to get better results. It's designed to help with some real hurdles in spotting early lung cancer, like when there's not much patient data available, or when we have very few examples of certain types of cases, or when our tools don't adapt well to new patients. Unlike the usual methods, our approach combines creating fake but useful data, putting multiple prediction models together, and then checking how sturdy and dependable the whole system is, all into one setup.

The system consists of five major stages:

- Data preprocessing and feature engineering
- Synthetic data generation using CTGAN
- Multi-model training and ensemble construction
- Comparative evaluation and ablation study
- Robustness and generalization assessment

FINAL ARCHITECTURE OF THE PROPOSED WORK



Data preprocessing and feature engineering

Clinical data often comes pretty messy, with missing pieces and lots of inconsistencies. That's why getting it ready, or "preprocessing," is really important for making sure our models actually work well and give reliable results.

Missing Values Handling

For handling missing values, we are employing a simple We fill in any missing numbers by using the average value. For missing categories, we just use the one that appears most often. We also add a simple yes or no signal to point out where data was originally missing, so that information isn't lost.

- Example 1: Missing "smoking history" → mode imputation + flag
- Example 2: Missing "age" → mean imputation + flag

Encoding Categorical Features

We change categories into numbers. If the categories don't have any particular order, like different colors, we use something called one-hot encoding. But for categories that do have an order, like 'small' to 'large,' we use label encoding. Doing it this way stops us from accidentally creating an order where there isn't one.

Normalization of Features

For normalizing our attributes, as they are on different scales, we are employing standardization. Hence, our numerical attributes are standardized as:

$$X_{norm} = X - \mu / \sigma$$

Here, μ and σ are feature-specific. Standardization helps our optimizer converge faster and produces more stable results for both tree-based and gradient-based models.

When we're done processing the data, we need a way to organize it so we can work with it. We basically set it up like a big table, and we've labeled this processed data as $X_{processed}$. In this table, 'n' tells us exactly how many individual samples we have. And 'd' refers to the number of specific clinical details, or features, that we're keeping track of for each of those samples.

The processed data set is written as:

$$X = (x_1, x_2, \dots, x_n), x_i \in R^d$$

Synthetic data generation using CTGAN

To deal with situations where we have an uneven number of items in different categories, we use something called a Conditional Tabular Generative Adversarial Network, or CTGAN. What CTGAN does is figure out how all the different parts of our data tables are connected. Then, it

creates new, realistic-looking examples of data, specifically for the categories we want to balance.

$$X_{synthetic} = G(z, c)$$

where:

- z = noise vector
- c = class condition
- G = generator network

Then, the dataset used for the classification task will be the combination of the processed dataset and the synthetic dataset:

$$X_{augmented} = X_{processed} \cup X_{synthetic}$$

Multi-model training and ensemble construction

A multi-model approach, which avoids depending upon just one classifier, is used for the following training process.

The models employed in this analysis include Random Forest and Gradient Boosting (XGBoost). Random Forest is effective in managing variance and addressing non-linearity within the data, while Gradient Boosting specifically focuses on reducing bias, enhancing the overall predictive performance of the model.

Each of the classifiers, RF and XG-Boost, predicts the probability:

$$P_i = f_i(X_{augmented})$$

where the function f_i denotes the i th classifier.

Then, the final probability is the weighted sum of the probabilities produced by the classifiers:

$$P_{final} = \sum_{i=1}^n w_i P_i$$

where w_i are the model weights used for the combination of the probabilities produced by the RF and XGBoost classifiers.

Comparative evaluation and ablation study

For an empirical verification of the efficacy of the proposed framework, a comparative analysis and an ablation study are performed. This is with the aim of evaluating the influence of each of these modules on the performance improvement, especially the influence of the CTGAN-based augmentation and the hybrid ensemble model.

The experiments involve four settings:

Baseline Setting: Using the original dataset without any augmentation

SMOTE Augmentation Setting: Conventional oversampling technique

CTGAN Augmentation Setting: Data augmentation with CTGAN without ensemble

Proposed Framework Setting: CTGAN + Hybrid Ensemble

This methodology facilitates the distinction between conventional approaches and the novel approaches, as well as determining the contribution of each of the modules. An ablation study involves analyzing the effect of excluding specific modules such as CTGAN or the hybrid ensemble in the system.

The training of the model involves a supervised learning approach where optimization techniques are applied through a categorical cross-entropy loss function defined as follows:

$$L = \frac{1}{K} \sum_{n=1}^N \sum_{k=1}^K y_{n,k} \log(\hat{y}_{n,k})$$

In this equation, N and K stand for the total number of samples and the number of output classes, respectively. $y_{n,k}$ and $\hat{y}_{n,k}$ refer to the true probability and the predicted probability.

In order to maximize performance and generalize better, hyperparameters optimization is applied systematically. For ensemble models, hyperparameters like the number of estimators, tree depths, and learning rates are selected based on grid and random search methods. Early stopping is also applied to the boosting model to avoid overfitting. Stratification sampling is conducted to balance the distribution of class among train-validation-test splits.

Furthermore, cross-validation is applied to test the robustness of the model among several data splits by reducing the effect of initialization.

Model Deployment and Clinical Decision Support

The final optimized ensemble model is implemented as a Flask-based application that serves as a clinical decision support system, facilitating prediction and user interaction. The implementation design is intended to provide efficient, scalable, and convenient usage in healthcare settings.

The application uses patient data as inputs, including their demographic information and symptoms, to run predictions through the pipeline and deliver predictive outputs. Specifically, the application delivers the following:

- Prediction of the probability of lung cancer risk
- Classification of lung cancer risk (low risk / high risk)
- Score of confidence in the prediction
- Feature contributions in making the decision

In contrast to regular non-interactive models, the application can be used interactively by healthcare practitioners, providing them with real-time insights and facilitating early detection of risks.

Experimental Results and Analysis

The experimentation with the developed enhanced predictive modeling technique was carried out using a clinical structured data set that included various patient characteristics related to the diagnosis of lung cancer. The variables considered in the study included demographic factors like age and gender, along with medical factors such as smoking status, coughing, fatigue, chest pain, and shortness of breath. The outcome variable under study was a binary classification model where low-risk and high-risk classes were distinguished.

To perform an unbiased assessment, stratified sampling was used to create three samples - the training set comprised 70% of the data, the validation set 15%, and the test set also accounted for 15% of the entire data set. Since it is common for clinical data sets to be class-imbalanced, such a sample preparation method helps preserve the distribution across the subsets. Additionally, for consistency reasons, each experiment was repeated five times.

In preparation for training the model, basic pre-processing methods such as imputation of missing values, encoding of categorical variables, and normalization of features were performed. Augmentation via CTGAN was done only on the training set to avoid leaking the test data set. This helps ensure that the model has no prior knowledge about the test data set.

Performance comparison between the proposed model and other baseline models, including Random Forest and Gradient Boosting, along with conventional augmentation approaches like SMOTE, was done based on various performance measures. These include Accuracy, Precision, Recall, F1-Score, and Area Under Curve (ROC).

Parameter	Value
CTGAN Optimizer	Adam ($\beta_1=0.5$, $\beta_2=0.9$)
Learning Rate	1×10^{-4} (Generator), 2×10^{-4} (Discriminator)
Batch Size	256
Epochs	250
Loss Function	Binary Cross-Entropy (GAN), Log Loss (Classifier)
Framework	Python (Scikit-learn, XGBoost, CTGAN Library)
Hardware Used	CPU (16 GB RAM) for inference, GPU (RTX 4090) for training
Data Split	70% Train / 15% Validation / 15% Test (Stratified)

Table 3 Training Configuration.

Evaluation metrics

Accuracy, precision, recall, F1-score, and ROC-AUC are essential metrics used to evaluate the performance of classification models. Accuracy measures the overall correctness of the model, while precision focuses on the proportion of true positive predictions among all positive predictions. Recall, on the other hand, assesses the model's ability to identify all relevant instances. The F1-score provides a balance between precision and recall, offering a single metric that reflects both aspects. Lastly, the ROC-AUC evaluates the model's ability to distinguish between classes across various threshold settings, providing insight into its overall performance.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

In terms of multi-class evaluations, the model's performance is measured based on the following: macro-average Precision, Recall, and F1-score. These measures allow for a fair assessment because all classes are considered equal irrespective of their prevalence.

Macro-averaged measures are defined as:

$$Precision_{macro} = \frac{1}{K} \sum_{k=1}^K Precision_k$$

$$Recall_{macro} = \frac{1}{K} \sum_{k=1}^K Recall_k$$

$$F1_{macro} = \frac{1}{K} \sum_{k=1}^K F1_k$$

where K denotes the total number of classes, and each measure is independently calculated for each class separately and then averaged. Such an approach is especially useful when working with imbalanced data, since it does not allow one class to dominate the others.

In order to test the discrimination power of the model, we compute the Receiver Operating Characteristic – Area Under the Curve (ROC-AUC). The area under the curve is an indicator of the power of the classifier in distinguishing between two classes.

The average time required to predict one observation is:

$$t_{infer} = T_{total}/N$$

where T_{total} is the overall time needed to make predictions (processing included), and N refers to the sample size. This

measure is essential in determining whether the model can be effectively utilized in real-world clinical settings.

In order to determine the statistical significance of the obtained results, a paired two-tailed t-test is performed through several experiments. The t-statistic is calculated as follows:

$$t \frac{-\bar{d}}{Sd / \sqrt{n}}$$

where $\bar{d}$ represents the is the average difference in performance between the models, Sd denotes the standard deviation of the differences, and n is the number of paired observations. Significance is evaluated at the 95% confidence level ($p < 0.05$), which guarantees that the improvements obtained are statistically significant and not due to random chance.

Interpretation

The experiment results clearly show that the introduced generative-augmented ensemble model outperforms baseline algorithms, such as random forest and gradient boosting, on all metrics used for the assessment. Namely, the use of CTGAN synthetic data augmentation solves the class imbalance issue, thus allowing better detection of the minority class examples, i.e., patients who are at high risk of the disease.

It has been noted that there was an increase in accuracy by about 3-5% in terms of macro F1-score. Such an improvement shows better precision and recall rates, which translates to better classification quality. For medical diagnosis, such improvement in terms of F1 score is important because it helps avoid false negatives in disease detection.

Additionally, the obtained results have been statistically validated, and it was found that they significantly differ from those produced by other models ($p < 0.05$). Moreover, the inference time is low (milliseconds), and the proposed model can be used in practice because its deployment uses Flask-based architecture.

Experimental Results Overview:

Evaluation of the generative model-based predictive framework has been performed using an experiment on structured data with patient characteristics indicative of the risk of developing lung cancer. Such characteristics include patient age, smoking history, coughing, chest pain, and others. There is a binary classification objective with values for low-risk and high-risk patients.

To guarantee reliable performance results without any bias due to data splitting, stratified sampling has been used to divide the dataset into training, validation, and testing sets by size of 70%, 15%, and 15%, respectively. Such division maintains the distribution among classes in each subset,

which is necessary when working with imbalanced medical datasets.

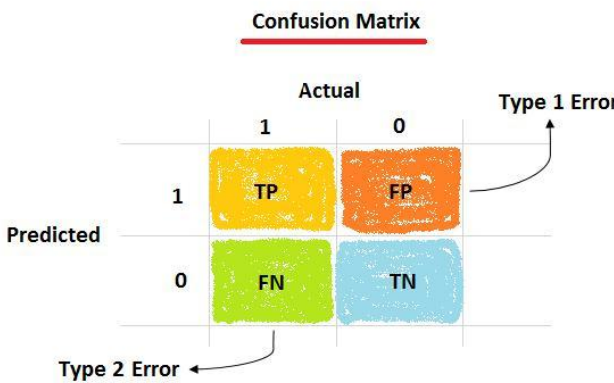
As part of the pre-processing, missing value filling, categorical variables transformation, and normalization have been performed. Augmentation using CTGAN has been implemented only on the training subset to avoid information leakage during evaluation. Based on the synthetic data generated in the process, Random Forest, XGBoost, and the novel ensemble model were trained.

Each experiment was conducted several times using different initializations, and their average values were used to minimize the impact of random variations. The performance of each model was measured using the usual classification metrics such as accuracy, precision, recall, F1-score, and area under the receiver operating characteristic curve (ROC-AUC).

Model	Accuracy (%)	Precision	Recall	F1 Score
Random Forest	91.2%	90.5%	88.7%	89.6%
XGBoost	93.5%	92.8%	91.2%	92.0%
SMOTE + RF	94.1%	93.0%	92.5%	92.7%
CTGAN + Ensemble	96.8%	95.9%	96.2%	96.0%

Table 5 Performance Metrics.

Confusion Matrix Analysis :



The confusion matrix is an essential concept used in the evaluation of performance of any binary classifier. Confusion Matrix is basically a 2x2 square matrix that consists of predicted and actual outputs. The confusion matrix includes four types of parameters, which include true positives (TP), false negatives (FN), true negatives (TN),

and false positives (FP). True positives are those instances where the output prediction of the classifier is a positive value, whereas true negatives refer to those instances where the model predicts negative results. These two metrics refer to cases where the model is accurate enough in terms of its prediction.

False positives, otherwise known as type I error, refer to instances where the output of the predictor model is positive, but in reality, the outcome is negative. A good example would be the misdiagnosis of patients, whereby the classifier predicts a disease in a patient who actually doesn't have one. In the same vein, false negatives, which are also referred to as type II errors, refer to instances where the predictor model fails to diagnose a disease when there really is one. A good example would be the presence of spam emails inside the inbox of a user.

Some of the common metrics that are calculated based on the Confusion Matrix are used to measure the effectiveness of a classifier model. The Accuracy metric is perhaps the most frequently used one, which represents the percentage of correct predictions compared to the total number of predictions. Accuracy, however, does not represent an optimal measure in some scenarios where there are many more instances from one class than from others, in which case predicting only one class yields high accuracy while being completely irrelevant.

Another two important metrics include Precision and Recall. While the former is an indication of how often the model predicts correctly the positives, the latter measures how often among all the positive cases predicted by the model, these are actually true positives. In this way, precision focuses on the minimization of the False Positives rate while Recall highlights the need for reducing the False Negatives rate.

The F1 score was designed to combine Precision and Recall into their harmonic average. This measure is most applicable when both False Positive and False Negative rates have substantial importance for the task. While accuracy is more appropriate to estimate a model's overall performance, the F1 score presents a better option for balanced assessments of the same. Nevertheless, even with the F1 score, one needs to consider the problem itself and the importance of certain aspects of the classification task.

In conclusion, the confusion matrix is not merely a way to evaluate model performance but an integral part of its evaluation and decision-making process. The relevance of certain metrics can vary from one field to another depending on the cost of making specific mistakes. In medicine, False Negative predictions have to be minimized, while in spam filters, it is False Positives that need to be avoided. Thus, to evaluate the models successfully, one must consider the implications of the metrics and the problem itself.

Conclusion and future scope:

This paper describes an advanced predictive model for lung cancer detection based on CTGAN and ensemble learning.

Combining synthetic data generation with the ensemble technique greatly boosts predictive power, especially in relation to the detection of the minority classes.

In contrast to the traditional method, the current research focuses on robustness and generalizability via systematic analysis, comprising ablation studies and noisy data tests. The conclusions prove that although synthetic data is a powerful tool, its use requires cautious validation.

Future efforts will concentrate on several key areas, including the validation of systems across multiple institutions to ensure robustness and reliability. Additionally, there will be a focus on the integration of imaging data through multimodal artificial intelligence, enhancing the depth of analysis. The application of explainable AI techniques, such as SHAP and LIME, will be prioritized to improve transparency and understanding of model decisions. Finally, optimizing deployment strategies for real-time healthcare systems will be essential to facilitate immediate and effective use in clinical settings.

REFERENCES

- [1] [1] B. Zhang, J. Rao, W. Jiang and Z. Gao, "Detection of Renal Cell Carcinoma Based on Clinical Data Using Machine Learning Approach," **2025 IEEE 5th International Conference on Soft-ware Engineering and Artificial Intelligence (SEAI)**, Fuzhou, China, 2025, pp. 36-40, doi: 10.1109/SEAI65851.2025.11108907.
- [2] A. Durairaj, R. K. M G, V. Bhat and A. H, "Renal Cell Carcino-ma (RCC) Staging Using Neural Networks: A Deep Learning Approach," **2025 International Conference on Biomedical En-gineering and Sustainable Healthcare (ICBMESH)**, Manipal, India, 2025, pp. 1-6, doi: 10.1109/ICBMESH66209.2025.11182260.
- [3] A. Kumar Chanchal, S. Lal and S. Suresh, "Development of Robust CNN Architecture for Grading and Classification of Re-nal Cell Carcinoma Histology Images," in **IEEE Access**, vol. 13, pp. 121849-121867, 2025, doi: 10.1109/ACCESS.2025.3586935.
- [4] J. Tao, A. McClelland, C. Wang and T. H. Zeng, "Early Diagnos-tic Method for Renal Cell Carcinoma Using Surface Enhanced Raman Spectroscopy," **2025 IEEE 20th Nanotechnology Mate-rials and Devices Conference (NMDC)**, Delhi, India, 2025, pp. 168-171, doi: 10.1109/NMDC64551.2025.11234018.
- [5] R. K, V. S. Kumari, N. Muthuvairavan Pillai, A. Rekha, M. Gun-asekaran and P. L. Mahimai Doss, "Machine Learning for Mul-ti-Class Kidney Tumour Segmentation: Advancing Classifica-tion Accuracy," **2025 5th International Conference on Perva-sive Computing and Social Networking (ICPCSN)**, Salem, India, 2025, pp. 601-607, doi: 10.1109/ICPCSN65854.2025.11035274.
- [6] M. K. Prakash, D. Ambika, S. R, P. Saravanan, N. Thangarasu and C. Kumuthini, "Kidney Cancer Diagnosis and Prognosis Using Ai and Machine Learning," **2025 9th International Conference on Inventive Systems and Control (ICISC)**, Coimbatore, India, 2025, pp. 663-668, doi: 10.1109/ICISC65841.2025.11188243.
- [7] S. I. Niloy, M. R. K. Khan, T. Mohaidat, Z. Niu, J. Eda and K. Khalil, "Machine Learning-Based Classification of Kidney Can-cer Subtypes Using MiRNA-Seq Data," **2025 IEEE 4th*

- International Conference on Computing and Machine Intelligence (ICMI)**, MI, USA, 2025, pp. 1-5, doi: 10.1109/ICMI65310.2025.11141163.
- [8] T. Shirdhone and S. D. Chede, "Performance Evaluation of CNN Models for Kidney Abnormality Detection: A Comparative Study," **2025 International Conference on Intelligent Computing and Knowledge Extraction (ICICKE)**, Bengaluru, India, 2025, pp. 1-6, doi: 10.1109/ICICKE65317.2025.11136743.
 - [9] X. Sun et al., "Engineering a Universal and Fully Automatic Analyzer for Multichannel and On-Site Biochemical Detection in Body Fluids," in **IEEE Transactions on Biomedical Engineering**, vol. 72, no. 1, pp. 177-186, Jan. 2025, doi: 10.1109/TBME.2024.3443626.
 - [10] J. Yu et al., "A Comprehensive Investigation to Identify Prognostic mRNA and miRNA Signatures for Renal Cell Carcinoma Utilizing a Stratification-Based Approach," in **Big Data Mining and Analytics**, vol. 7, no. 4, pp. 1396-1416, December 2024, doi: 10.26599/BDMA.2024.9020039.
 - [11] H. R. D. PP, K. S. and S. Umapathy, "Automated Segmentation of Renal Cell Carcinoma in CT Images using Deep Learning Techniques," **2024 International Conference on Distributed Systems, Computer Networks and Cybersecurity (ICDSCNC)**, Bengaluru, India, 2024, pp. 1-6, doi: 10.1109/ICDSCNC62492.2024.10939853.
 - [12] J. Niveditha, S. Supreeth and K. Patil, "Renal Cell Carcinoma Classification: Deep Learning with MLflow, DVC, and AWS CI/CD Deployment," **2024 8th International Conference on Electronics, Communication and Aerospace Technology (ICECA)**, Coimbatore, India, 2024, pp. 634-641, doi: 10.1109/ICECA63461.2024.10800992.
 - [13] M. A. Aziz et al., "Deep Learning Approach for Renal Cell Carcinoma Detection, Subtyping, And Grading," **2024 IEEE International Conference on Image Processing (ICIP)**, Abu Dhabi, United Arab Emirates, 2024, pp. 3861-3867, doi: 10.1109/ICIP51287.2024.10647236.
 - [14] G. Kaur, N. Sharma, S. Malhotra, S. Devlival and R. Gupta, "Kidney Tumor Detection and Classification Using Convolutional Neural Network Architecture," **2024 2nd World Conference on Communication & Computing (WCONF)**, RAIPUR, India, 2024, pp. 1-5, doi: 10.1109/WCONF61366.2024.10691954.
 - [15] Z. Ye, S. Ge, M. Yang, C. Du and F. Ma, "An Explainable Classification Model of Renal Cancer Subtype Using Deep Learning," **2024 17th International Congress on Image and Signal Processing, BioMedical Engineering and Informatics (CISP-BMEI)**, Shanghai, China, 2024, pp. 1-10, doi: 10.1109/CISP-BMEI64163.2024.10906194.
 - [16] J. Wan, Y. Hao, X. Ma and H. Wang, "Multi-Scale Graph-Based Cross-Attention Transformer for Whole Slide Image Classification," **2024 18th International Conference on Control, Automation, Robotics and Vision (ICARCV)**, Dubai, United Arab Emirates, 2024, pp. 1-6, doi: 10.1109/ICARCV63323.2024.10918883.
 - [17] R. Chauhan, M. Karnati and P. Singh, "Attention Based Deep Neural Network for Classification of Kidney Ailments Using CT Images," **2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT)**, Kamand, India, 2024, pp. 1-5, doi: 10.1109/ICCCNT61001.2024.10723909.
 - [18] N. Nithya Sree and R. Ramyadevi, "Multiple Cancer Prediction with Image Analysis and Machine Learning," **2024 IEEE International Conference on Computing, Power and Communication Technologies (IC2PCT)**, Greater Noida, India, 2024, pp. 481-485, doi: 10.1109/IC2PCT60090.2024.10486632.
 - [19] M. T. A. Bhuiyan, M. N. H. Bhuiyan, K. M. M. Uddin and M. A. Based, "A Feature Selection-Based Ensemble Machine Learning Method for Predicting Chronic Kidney Cancer," **2024 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON)**, Dhaka, Bangladesh, 2024, pp. 193-199, doi: 10.1109/BECITHCON64160.2024.10962730.
 - [20] S. Verma and J. Arora, "Early Detection of Lung Cancer Using Machine Learning Techniques," **2024 13th International Conference on System Modeling & Advancement in Research Trends (SMART)**, Moradabad, India, 2024, pp. 846-850, doi: 10.1109/SMART63812.2024.10882179.
 - [21] S. Katariyar, P. Tendolkar, S. Chavan, S. Krishnan, S. Shinde and N. Sable, "Survival prediction in renal cell carcinoma patients using machine learning," **2023 6th International Conference on Advances in Science and Technology (ICAST)**, Mumbai, India, 2023, pp. 362-367, doi: 10.1109/ICAST59062.2023.10455029.
 - [22] S. Katariyar, P. Tendolkar, S. Chavan, S. Krishnan, S. Shinde and N. Sable, "Survival prediction in renal cell carcinoma patients using machine learning," **2023 6th International Conference on Advances in Science and Technology (ICAST)**, Mumbai, India, 2023, pp. 362-367, doi: 10.1109/TCBB.2022.3214190.
 - [23] T. Thakur, I. Batra, A. Malik, D. Ghimire, S. -H. Kim and A. S. M. Sanwar Hosen, "RNN-CNN Based Cancer Prediction Model for Gene Expression," in **IEEE Access**, vol. 11, pp. 131024-131044, 2023, doi: 10.1109/ACCESS.2023.3332479.
 - [24] K. S. Gill, V. Anand and R. Gupta, "Kidney Disease Classification Using Machine Learning Approach on DenseNet201 Model using Xray Images," **2023 First International Conference on Advances in Electrical, Electronics and Computational Intelligence (ICAEECI)**, Tiruchengode, India, 2023, pp. 1-4, doi: 10.1109/ICAEECI58247.2023.10370846.
 - [25] Kommalapati Naveen Kumar, P. Dhanalakshmi, K.K. Baseer. (2025). A Novel Transfer Learning Framework for Detection and Identification of Brain Tumors in Smart Healthcare Ecosystem. *Journal of Applied Bioanalysis*, 11(8), 461-479. <https://doi.org/10.53555/jab.v11i8.1960>